

STAT 412 HW07 — Concise Answer Key

1. Type I fill-ins: 31% or more; less than 31%. Type II fill-ins: less than 31%; 31% or more.
2. $\alpha = 0.0609$. For $p = 0.3$, $\beta = 0.8695$; for $p = 0.5$, $\beta = 0.8454$. Not a good test: α is relatively small, but at least one β is relatively large.
3. (a) $\alpha = 0.0718$. (b) When $\mu=195$, $\beta = 0.1151$.
4. Choice E: $H_0: \mu=41$; $H_a: \mu<41$. $t=-2.43$; exact P-value about 0.009 (table range $0.0075<P<0.01$). Reject H_0 ; sufficient evidence that $\mu<41$.
5. $z = -2.95$; P-value = 0.003. Reject H_0 ; there is sufficient evidence that the mean lifetime is not equal to 800 hours. Reject for significance levels greater than the P-value.
6. $H_0: \mu=162.8$; $H_a: \mu \neq 162.8$. $z=3.06$; two-tailed P-value = 0.002 to three decimals (exact 0.0022389), so reject H_0 . Reject for significance levels greater than the P-value.
7. Minimum sample size $n=41$.
8. Minimum sample size per group $n=26$.
9. Choice C: $H_0: p=0.4$; $H_a: p \neq 0.4$. Test statistic/model: X Binomial($n=20, p=0.4$). MyStatLab two-sided exact P-value = $2P(X \geq 12) = 0.1131$; do not reject H_0 at $\alpha=0.05$. Conclusion choice B.
10. chi-square = 50.93; P-value = 0.718. Not significant; fail to reject H_0 ; insufficient evidence that sigma squared differs from 4.1.
11. $H_0: \mu=10$; $H_1: \mu>10$ (equals for H_0 , greater-than for H_1). $t=3.49$; right-tail P-value about 0.0003 (enter 0.000 to three decimals), so reject H_0 . Reject for significance levels greater than the P-value.
12. Choice D: $H_0: \mu=18000$; $H_a: \mu>18000$. The screenshot mean is 18,330, so $t=0.94$; exact right-tail P-value about 0.176 (table range $0.15<P<0.20$). Do not reject H_0 ; there is not sufficient evidence. Reject for significance levels greater than the P-value.
13. Choice D: $H_0: p=0.097$; $H_a: p<0.097$. $z=-0.30$; P-value = 0.3821; do not reject H_0 . Conclusion choice C.
14. Choice D: $H_0: p=0.5$; $H_a: p<0.5$. $z=-5.09$; P-value < 0.0001 , so reject H_0 . Conclusion choice A.
15. Choice C: $H_0: p=0.54$; $H_a: p \neq 0.54$. $z=1.19$; P-value about 0.235, so do not reject H_0 from the visible 55.0% data.
16. Choice D: $H_0: p=0.5$; $H_a: p>0.5$. $z=21.17$; P-value essentially 0, so reject H_0 .
17. P-value = 0.0139; reject H_0 at $\alpha=0.10$; the alternative is supported.
18. P-value = 0.0093; reject H_0 at $\alpha=0.01$; the alternative is supported. Conclusion choice B.
19. Choice C.
20. Choice B.